

Program overview

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Year 2024/2025
Organization Mechanical Engineering
Education Minors Biomedical Engineering

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
BME-MI-087-24	BME-MI-087-24 Biomedical Engineering		
BM3102A	Bewegingsapparaat: anatomie en sturing	4	
BM3105-18	BME Minor Projects	14	
BM3106-18	Orthopedic Biomaterials	4	
BM3108-23	Medical Imaging	5	
BM3110	General Introduction to Biomedical Engineering	3	

Year	2024/2025
Organization	Mechanical Engineering
Education	Minors Biomedical Engineering

BME-MI-087-24

Contact for students	Dr.Ir. Aimée Sakes +31 27 84230 a.sakes@tudelft.nl
Prerequisites	Completion of the P-exam is strongly advised
Program Goals	At the end of this minor, the students will be able to: - describe the structure and the function of the musculoskeletal system in the human body from an engineering perspective. - describe the characteristics of the fields of application of medical devices and imaging modalities, as well as the end-users in these fields. - understand the collaboration complexity between clinicians and engineers in the development process and use of medical-technical equipment and systems. - explain the physical and engineering principles involved in specific medical devices and imaging modalities used in medicine. - describe typical phases in the development process of medical devices. - answer clinical questions based on biomedical research or medical designs using a scientific approach.
Administration by the Faculty of	3mE
Administration by the Education of	BME
Minor Title	Minor Biomedical Engineering
Intended for	BSc WB & BSc MT (not BSc KT)
Expected prior Knowledge	Basic knowledge of mechanics, Fourier-analysis and Matlab
Minor Exit Qualifications	The minor provides insight into the biomedical technology, the interaction with the patient and the use of technology in medicine.
Minor Coherence / Goal	The minor discusses biomedical technologies with a focus on: - the human musculo-skeletal system - minimally invasive techniques - imaging techniques, e.g. X-ray, CT, MRI, and Ultrasound Guest lectures by physicians provide a strong focus on the medical discipline. Recent developments will be discussed based on research within the Medical Delta, a cooperation between the Delft University of Technology, the Leiden University Medical Center, and the Erasmus Medical Center [www.medicaldelta.nl].
Minor Content 1	The minor comprises 5 parts: BM3101 General Introduction (4EC) BM3102A Human musculo-skeletal system: anatomy & control (4EC) BM3106-18 Orthopedic Biomaterials (4EC) BM3108 Imaging Techniques (4EC) BM3105-18 Assignment (14EC)
Education Methods	Lectures, assignments. Location: Delft with guest lecturers from Leiden and Rotterdam. Excursions
Minor Assessment	See course descriptions
Maximum number of participants	80
Minimum number of participants	30

BM3102A	Bewegingsapparaat: anatomie en sturing	4
Responsible Instructor	Prof.dr. H.E.J. Veeger	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Lectures: Anatomy & Control Basics of Human Anatomy Basics of biomechanics: application of simple statics to analyse motion modelling human motion and the musculoskeletal system structure - function relationship in normal and pathological conditions (haptic) feedback systems Measurement techniques	
Study Goals	To acquire: - basic knowledge of the human musculoskeletal system, in particular the anatomy and syndesmology of the extremities; - basic knowledge of the human nervous system, reflex functions and sensory modalities To be able to: - apply (basic) mechanics to understand human motion - construct simple free body diagrams visualising joint- and segment forces and moments - distinguish between causative and compensatory mechanisms in human motion - develop adequate hypothesis on muscle function and dysfunction based on motion behaviour Be familiar with: - clinical application of the above knowledge and skills	
Education Method	semester course (22 weeks): lectures and assignments.	
Books	Chapters from Human Anatomy & Physiology (10th Edition) (Marieb, Human Anatomy & Physiology) chapters from: Routledge Handbook of Biomechanics and Human Movement Science (available as e-book) selected journal publications	
Assessment	Electronic Exam Assignment(s)	
Elective	Yes	
Department	ME Department Biomechanical Engineering	
Contact	h.e.j.veeger@tudelft.nl	

BM3105-18	BME Minor Projects	14
Responsible Instructor	Dr. A. Sakes	
Instructor	Prof.dr. J.J. van den Dobbelaars	
Contact Hours / Week x/x/x/x	x/x/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	This assignment is the final project linked to this minor.	
Study Goals	Students will work on research- and/or design assignments within the field of biomedical engineering. Depending on the availability of assignments and student number, group size is not predetermined but usually will be four students. Teams of students with different backgrounds will be formed. Starting from a given research question, students formulate a research plan, do the research / design and present their results in the form of a short paper and an oral presentation during the closing symposium. At the end of this course, the students will be able to: - answer clinical questions based on biomedical research or medical designs using a scientific approach. - classify and identify different research designs. - carry out independent research in a responsible manner. - present your results in a scientific format, as well as on a clinically understandable level.	
Education Method	Project	
Assessment	Report and presentation (end of January)	
Department	ME Department Biomechanical Engineering	

BM3106-18	Orthopedic Biomaterials	4
Responsible Instructor	M.J. Mirzaali Mazandarani	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course is designed to equip students with the fundamental knowledge of characterisation techniques used for the evaluation of biological materials (e.g., bone). It also deals with the methods in the design, characterization, fabrication, and testing of orthopaedic biomaterials. This knowledge will help students to become familiar with the requirements necessary in the design of orthopaedic biomaterials. They will also be introduced to the advanced fabrication technologies (e.g., additive manufacturing techniques, AM, also known as 3D printing) which includes steps for processing and post-processing required for the fabrication of 3D printed biomaterials.	
Study Goals	At the end of this course, the students should be able to: i) explain various characteristic techniques (from nano-macro) used for the characterisation of hard biological materials ii) explain fundamental key design elements in the structure of natural materials iii) describe the property-structure relation of porous biomaterials iv) delineate essential parameters involved in the design of (porous or biomimetic) biomaterials v) critically discuss existing 3D-printing technologies, their principles, advantages, and limitations vi) critically examine the data evaluation methods on the characterisation of hard biological materials	
Education Method	Lectures and workshop sessions	
Literature and Study Materials	Lecture slides and recommended literature	
Assessment	Written Online Exam. If the opportunity arises for on-campus examinations, the prescribed method of testing will be: Written Exam on Campus. 3 group assignments also need to be delivered.	
Department	ME Department Biomechanical Engineering	

BM3108-23	Medical Imaging	5
Responsible Instructor	Dr. S.D. Weingärtner	
Instructor	Q. Tao	
Contact Hours / Week x/x/x/x	4(HC)1(S).0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Second year level in physics or engineering studies. Examples include Applied Physics, Mechanical Engineering, Electrical Engineering.	
Course Contents	A thorough mathematical basis is essential to follow the class. This can for example include Linear Algebra I + II and Calculus I + II. An in-depth understanding of Fourier analysis and basic physical understanding of electromagnetism is necessary. This course is part of the minor Biomedical Engineering. Class Medical Imaging Techniques: 5 ECTS, 1st quarter. In this class the most important methods and techniques for medical imaging will be introduced. We will discuss multiple imaging modalities and an introduction to medical image analysis and processing. Specifically the class will focus on the following content: 1. Ultrasound 2. X-Ray / CT 3. Positron emission tomography (PET) / Single-photon emission computed tomography (SPECT) 4. Magnetic Resonance Imaging (MRI) 5. Medical Image Analysis / Processing	
Study Goals	1. Assess the concepts and properties of medical images created by radioactivity in nuclear medicine, x-rays in X-Ray and CT, and sound waves in ultrasound 2. Discuss what creates a signal in magnetic resonance imaging and how this can be turned into an image 3. Contrast the different imaging modalities in terms of the underlying concepts, medical image information, and clinical applications 4. Apply image processing techniques to enhance image quality and extract relevant image features for analysis	
Education Method	The class will be composed of two parts: Lectures and discussion sessions. 1. Lectures: Frontal teaching in form of 10 lectures, 2 hours each, will be held across six weeks of the semester. 2. Discussion sessions: Throughout the semester discussion sessions will be offered. These sessions serve to recap of the most important or most challenging lecture material, provide an in-depth perspective, and discuss solutions to problem sets that are handed out after the lectures. The problem sets will not be graded. However, to be eligible to participate in the exam a student needs to present their solution to at least one problem in one of the discussion groups. A detailed schedule of lectures and discussion sessions will be made available at the beginning of the semester.	
Assessment	A written exam at the end of the year will be part of the class. The final grade will be determined by the result of the written exam.	

BM3110	General Introduction to Biomedical Engineering	3
Responsible Instructor	Dr. A. Sakes	
Contact Hours / Week x/x/x/x	4(HC).0.0.0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Course Contents	In this course, introductory lectures are given on the field of biomedical engineering and the role of technology in the advancement of healthcare. Topics include: - Medical Device Design - Biomaterials - Public Health(care) - Biomechanics	
Study Goals	At the end of this course, you will: - Have insight into the scope and subdisciplines within the field of biomedical engineering - Be able to identify stakeholders in the field of biomedical engineering. - Understand the collaboration complexity between clinicians and engineers in the development process and use of medical-technical equipment and systems. - Have insight in the history of the biomedical engineering field.	
Education Method	Lectures, self-study, and discussions. Self-arranged excursion to Museum Boerhaave in Leiden.	
Assessment	There are two written assignments that need to be completed in the course. The first assignment is an essay on a medical innovation. The second assignment is a group assignment in which you work on a design topic in sustainable biomedical engineering.	

Prof.dr. J.J. van den Dobbelsteen

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